

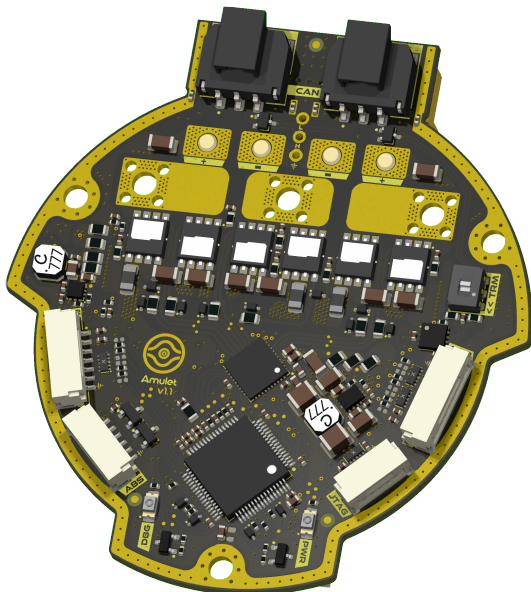
Amulet Motion Controller

Variant: Released

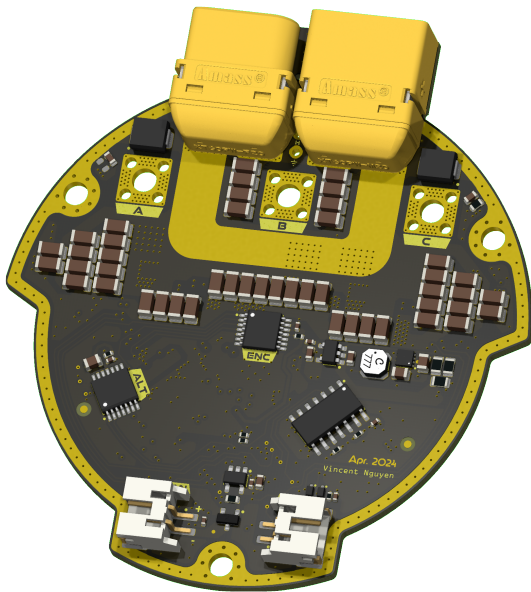
2024-11-14
Rev 1.1

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TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

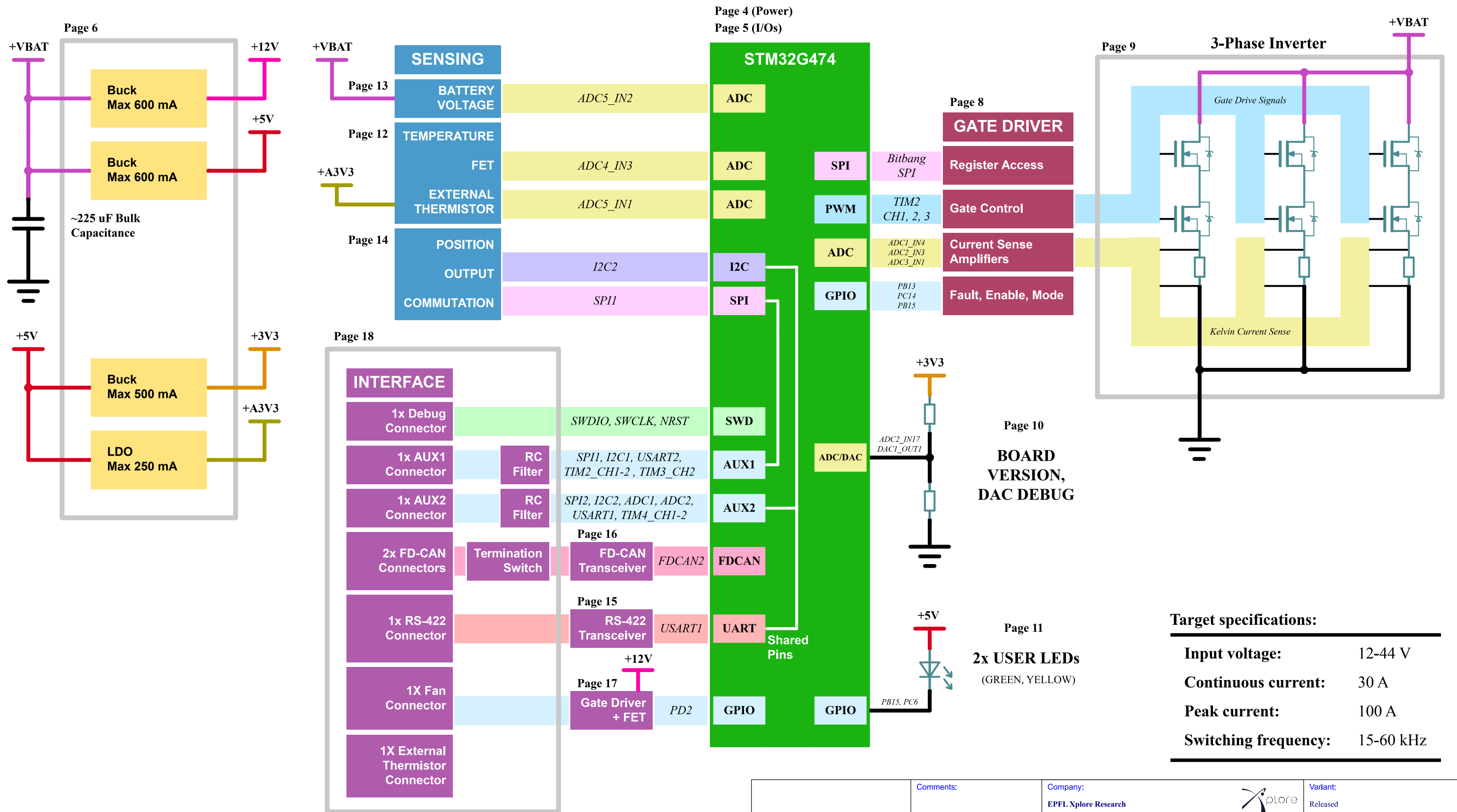
NOTES

- Schematic based off Josh Pieper's moteus controllers.
- Not fitted components are marked as **X**
- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

Released 13-APR-2024

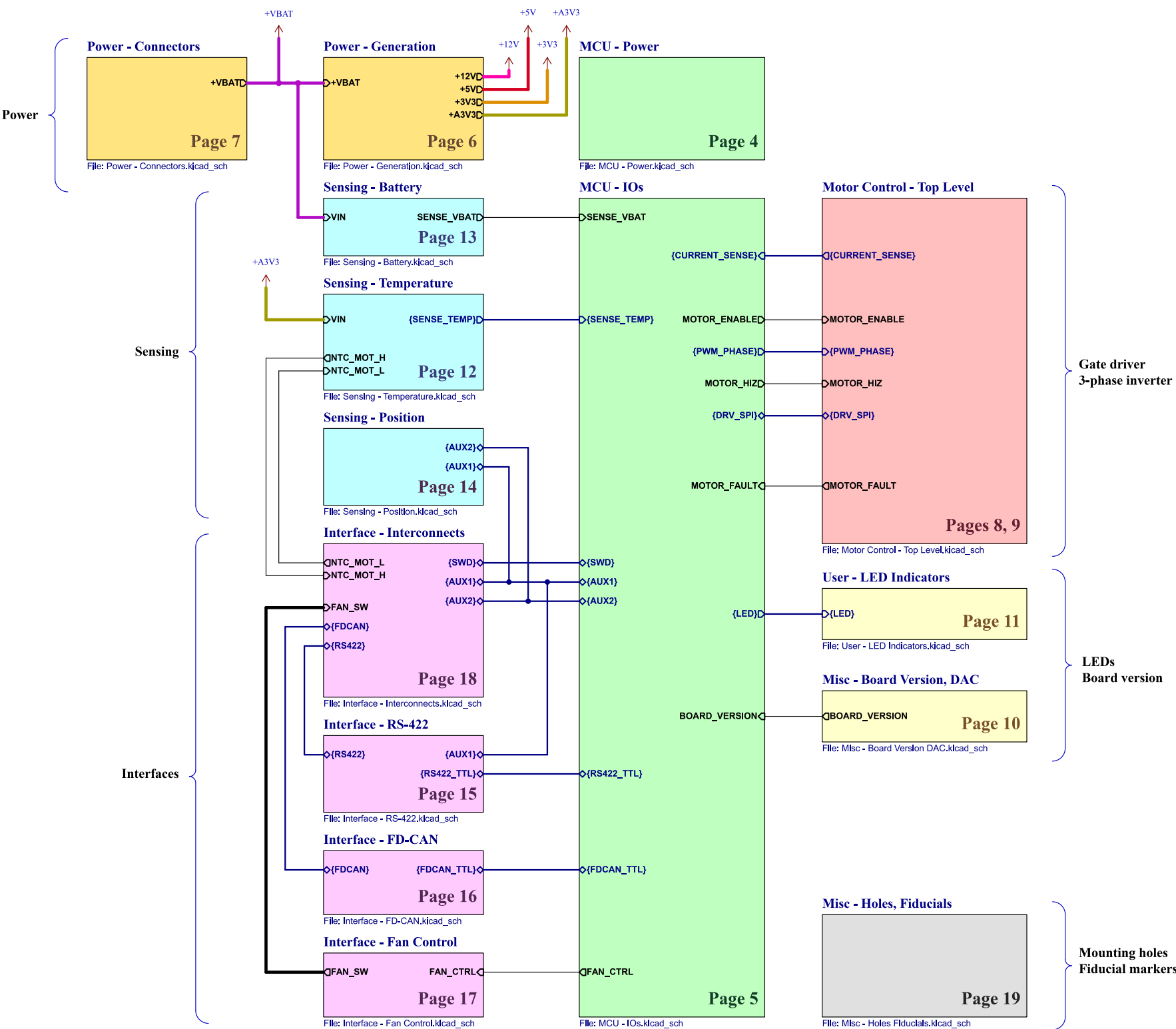
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	Sheet Title: Cover Page	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Path: /	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13
			Reviewer:	Revision: 1.1
			Size: A3	Sheet: 1 of 21

[2] Block Diagram



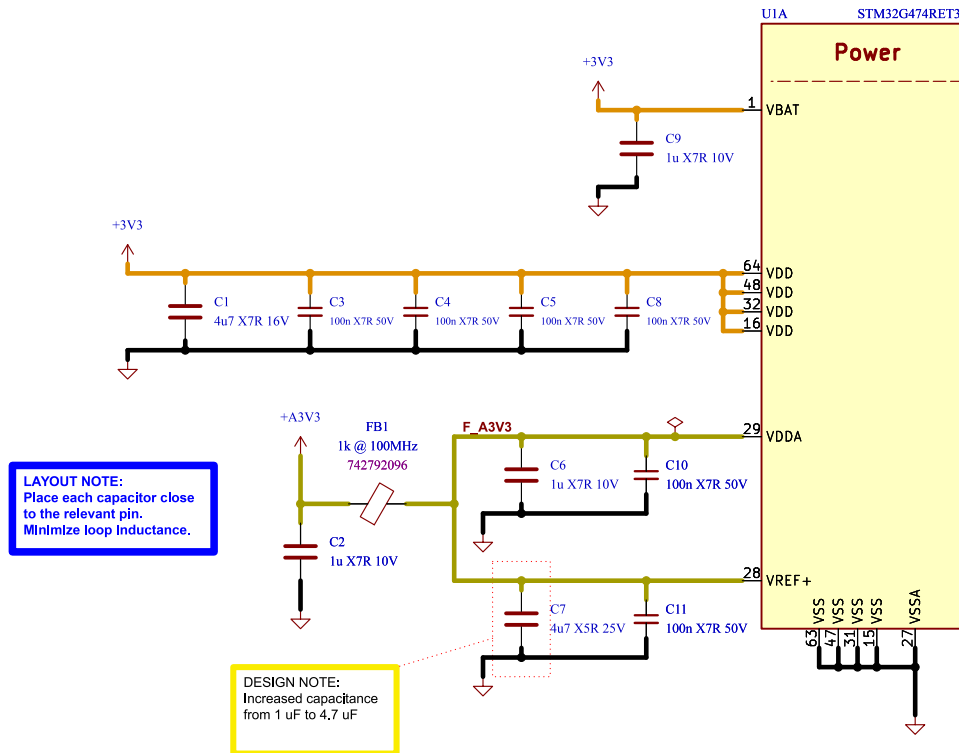
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Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 21	

[3] Project Architecture



Comments:	Company: EPFL Xplore Research				Variant: Released	
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1		
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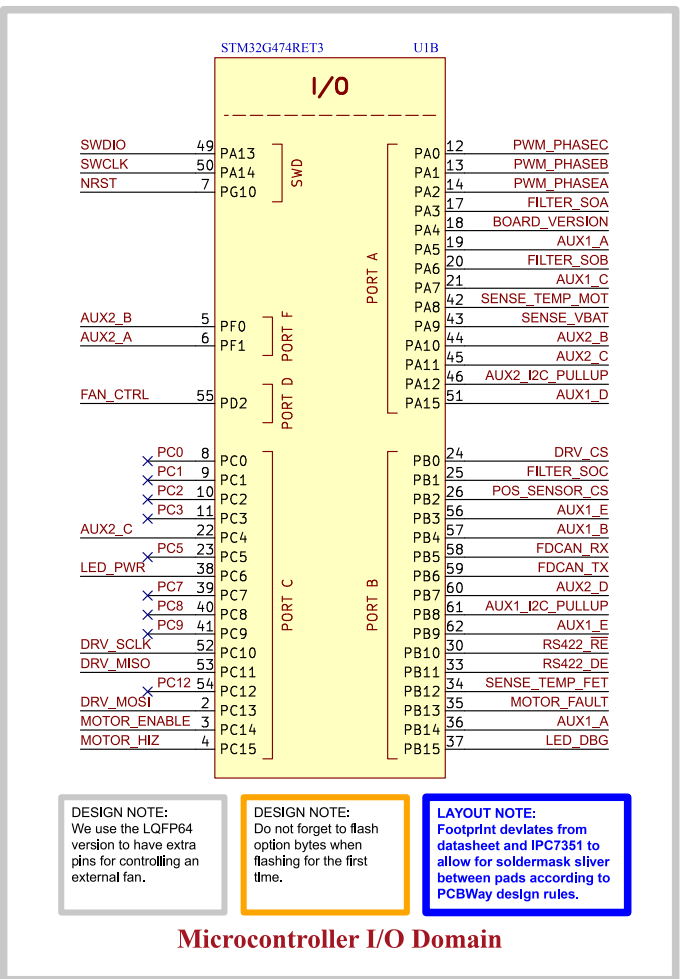
[4] MCU - Power



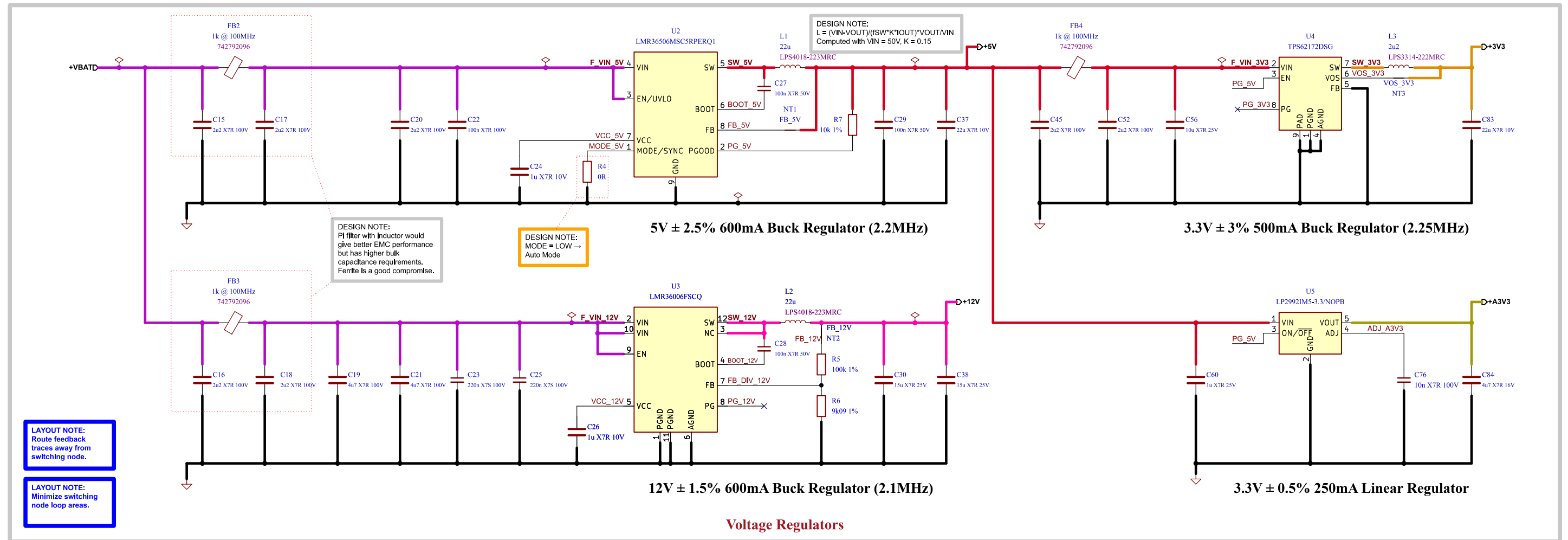
Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation		Company: EPFL Xplore Research		 Variant: Released				
			Board Name: Amulet Motion Controller		Project Name: Chienpanzé				
	Sheet Title: MCU - Power		File Name: MCU - Power.kicad_sch		Designer: Vincent Nguyen				
				Date: 2023-12-18		Revision: 1.1			
Sheet Path: /Project Architecture/MCU - Power/				Reviewer:		Size: A4		Sheet: 4 of 21	

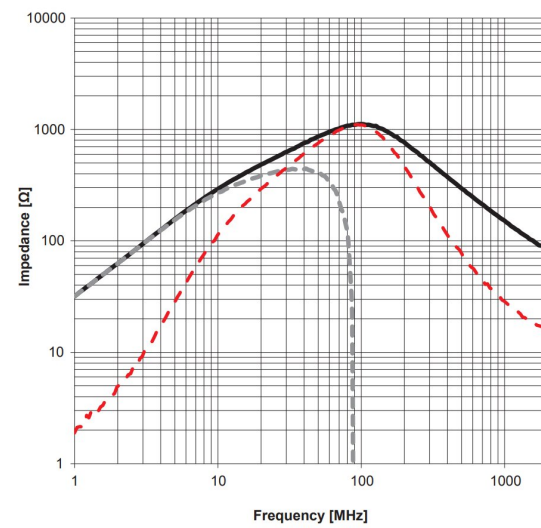
[5] MCU - I/Os



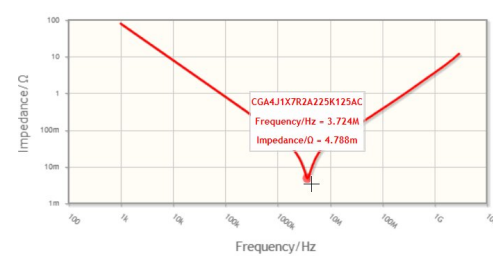
[6] Power - Generation



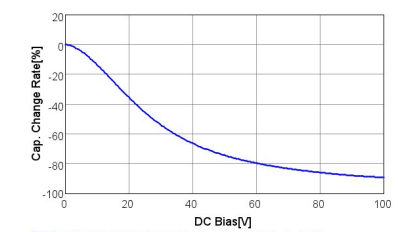
Ferrite bead Impedance vs Frequency curve



Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Capacitor and Ferrite Bead Parameters

Battery

+VBAT TP39 +VBAT
+VBAT TP40 +VBAT
+VBAT TP41 +VBAT

Filtered Power Rails

F_VIN_12V TP42 F_VIN_12V
F_VIN_5V TP43 F_VIN_5V
F_VIN_3V3 TP44 F_VIN_3V3

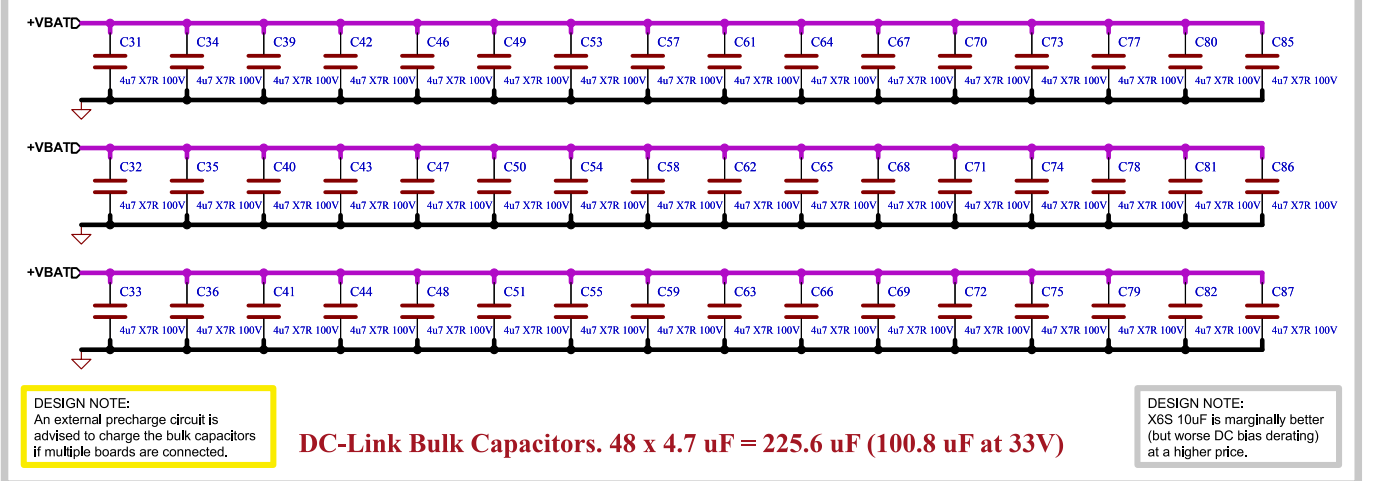
Power Rails

+12V TP45 +12V
+5V TP46 +5V
+5V TP47 +5V
+3V3 TP48 +3V3
+3V3 TP49 +3V3
+3V3 TP50 +3V3
+A3V3 TP51 +A3V3

Ground

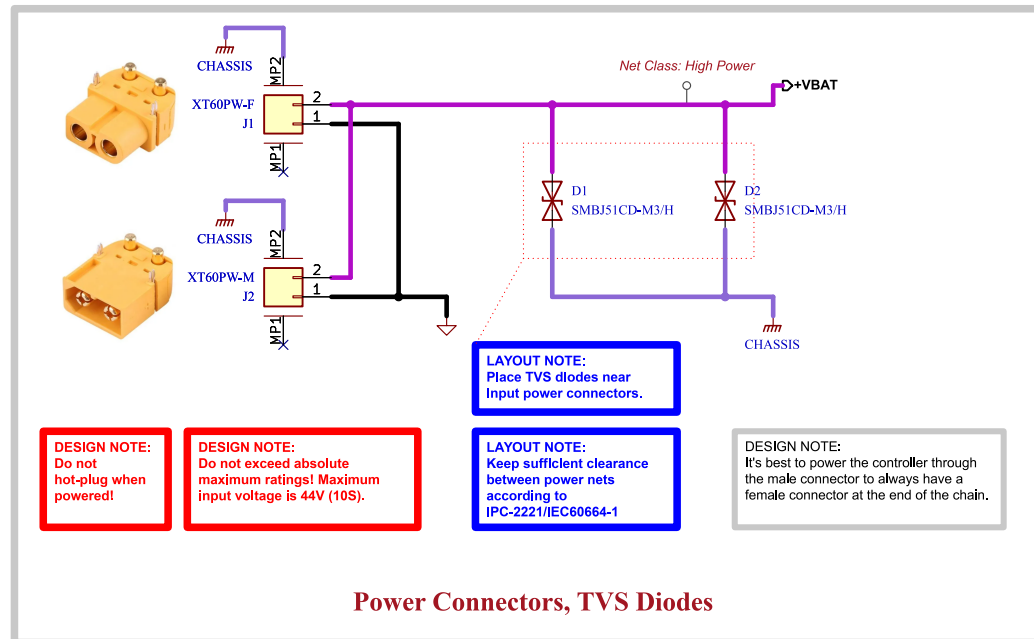
GND TP52 GND
GND TP53 GND
GND TP54 GND
GND TP55 GND
GND TP56 GND
GND TP57 GND
GND TP58 GND

Test Points



Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V		Company: EPFL Xplore Research		Variant: Released	
Sheet Title: Power - Generation		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Path: /Project Architecture/Power - Generation/		File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
		Reviewer:		Size: A3	Sheet: 6 of 21

[7] Power - Connectors



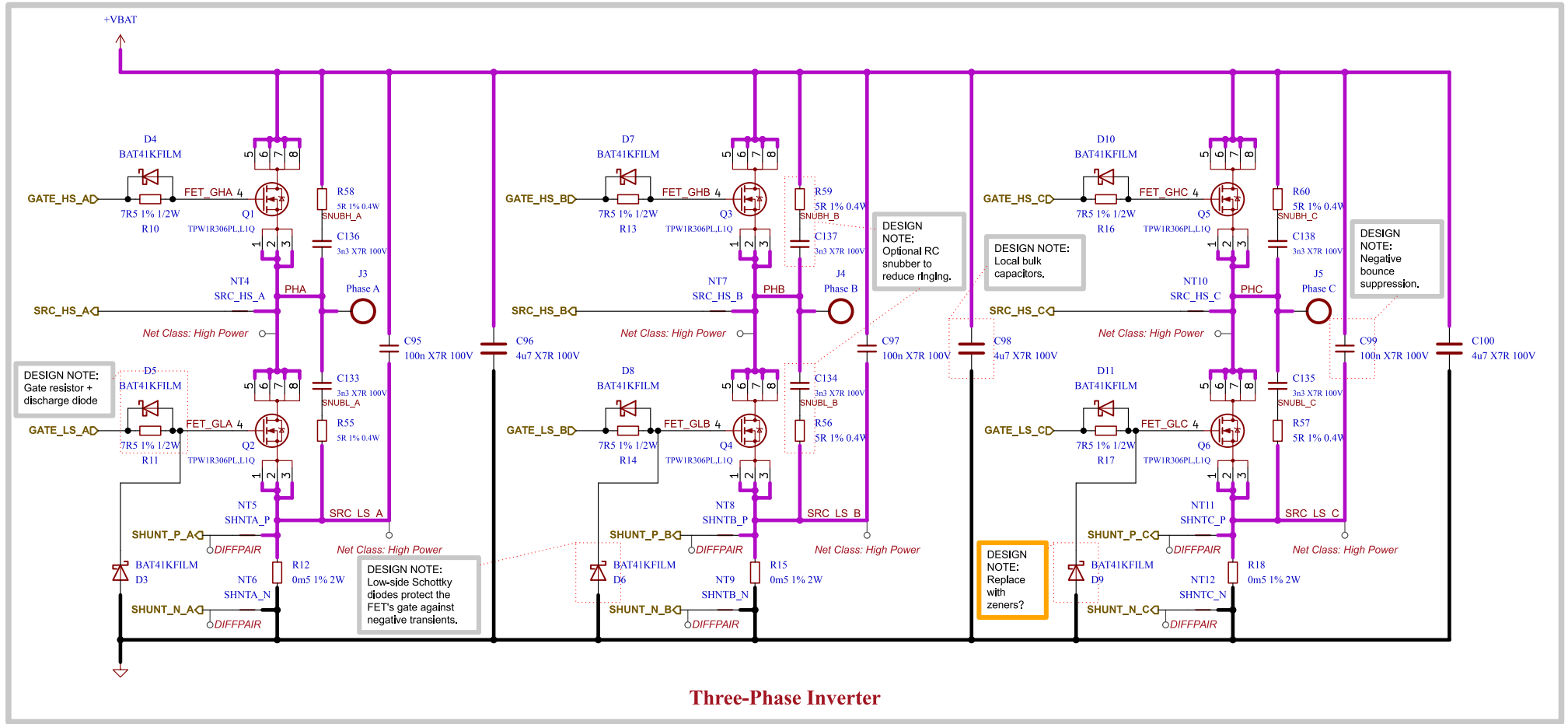
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100-V Three-Phase Smart Gate Driver



	Comments:	Company: EPFL Xplore Research			Variant: Released	
		Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
Sheet Title: Motor Control - Top Level	File Name: Motor Control - Top Level.kicad_sch	Designer: Vincent Nguyen		Date: 2023-12-20	Revision: 1.1	
Sheet Path: /Project Architecture/Motor Control - Top Level/		Reviewer:		Size: A3	Sheet: 8 of 21	

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:
System Design Considerations for High-Power Motor Driver Applications
Best Practices for Board Layout of Motor Drivers
Proper RC Snubber Design for Motor Drivers

Sheet Title:
Motor Control - Inverter

Sheet Path:
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:
EPFL Xplore Research

Board Name:
Amulet Motion Controller

File Name:
Motor Control - Inverter.kicad_sch

Reviewer:



Variant:
Released

Project Name:
Chienpanzé

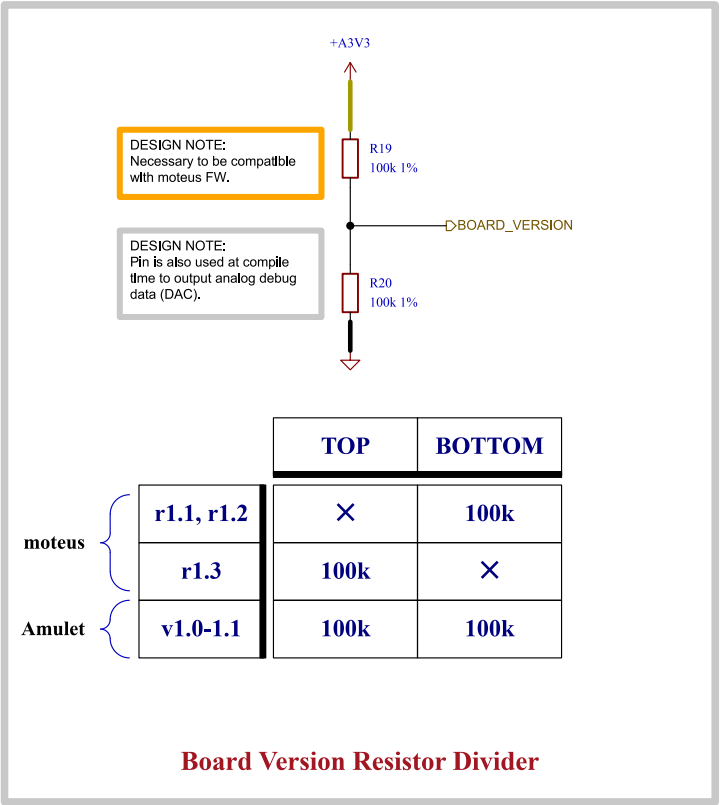
Date:
2024-01-25

Size:
A4

Revision:
1.1

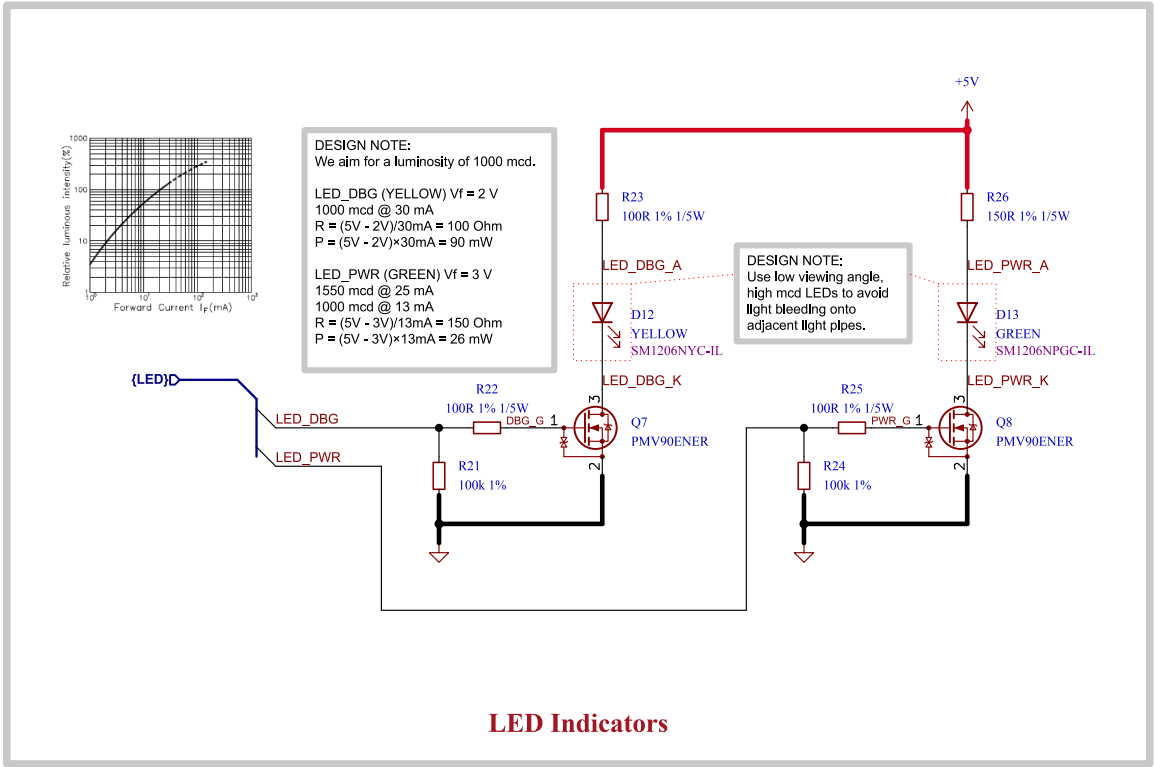
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[10] Misc - Board Version, DAC



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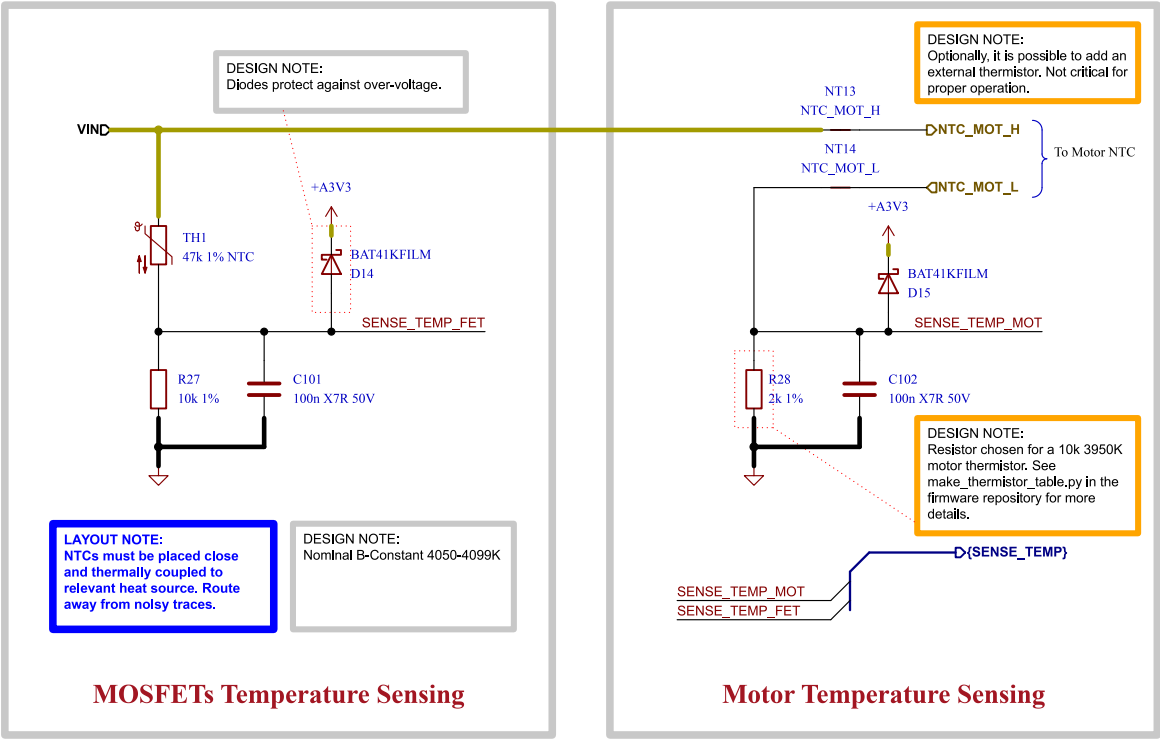
[11] User - LED Indicators



LED Indicators

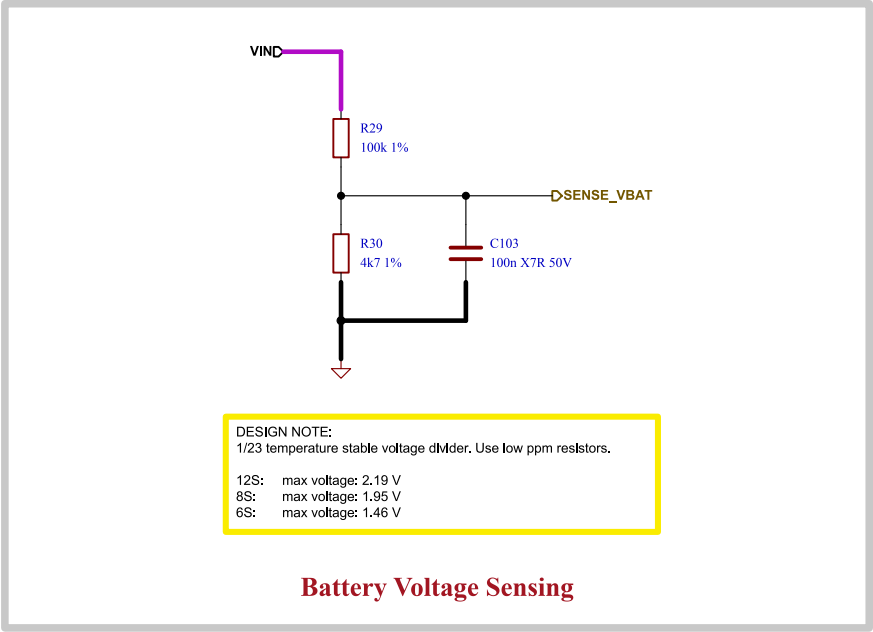
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[12] Sensing - Temperature



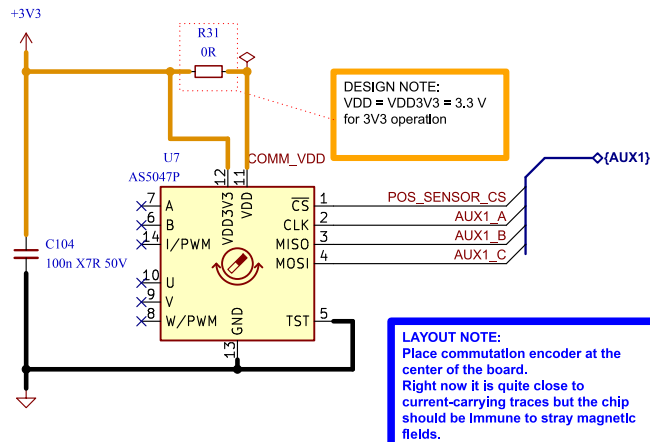
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[13] Sensing - Battery



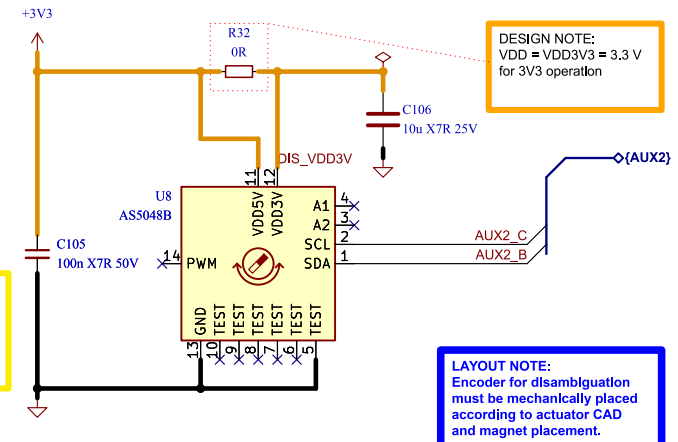
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	Sheet Path: /Project Architecture/Sensing - Battery/			Reviewer:	Size: A4	Sheet: 13 of 21

[14] Sensing - Position



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)

DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.



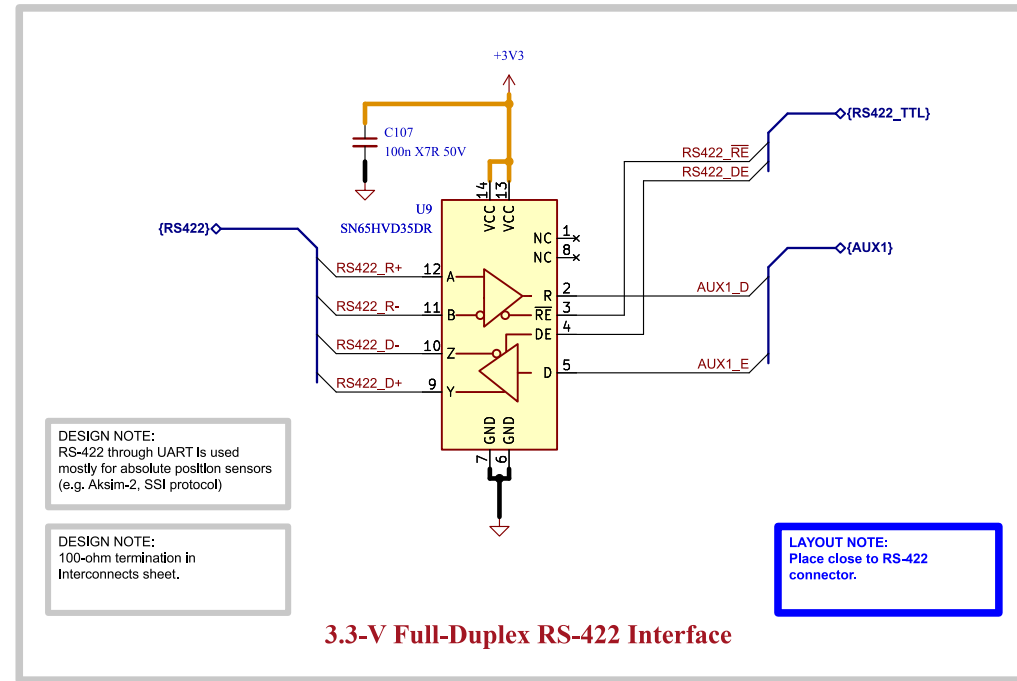
14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)

DESIGN NOTE:
The encoder for disambiguation cannot be used to estimate backlash!

LAYOUT NOTE:
Encoder for disambiguation must be mechanically placed according to actuator CAD and magnet placement.

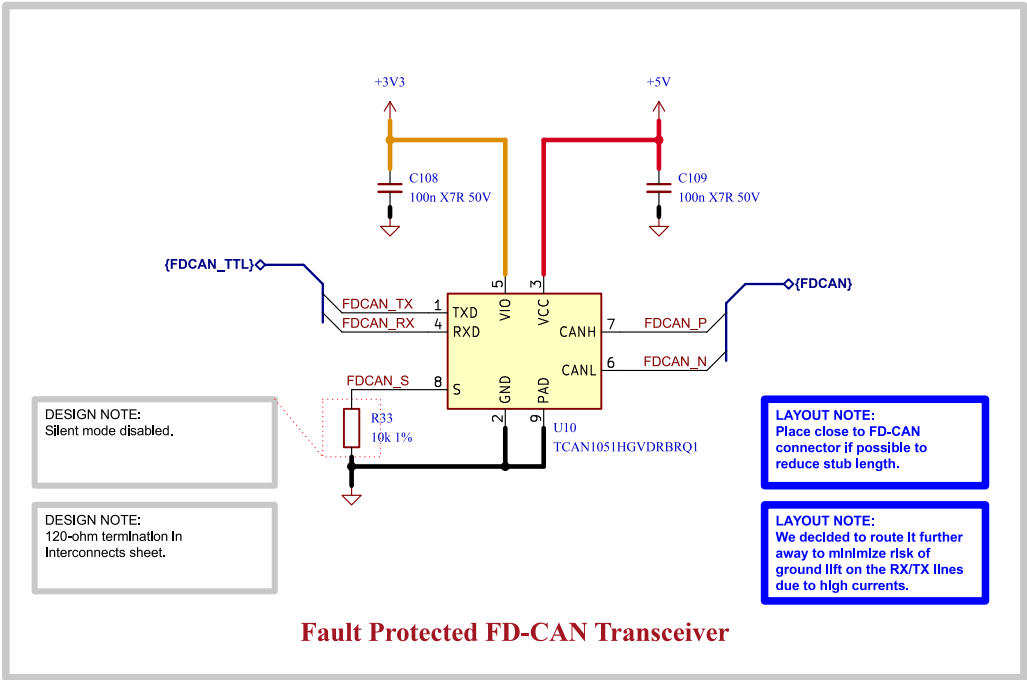
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			Sheet: 14 of 21	

[15] Interface - RS-422



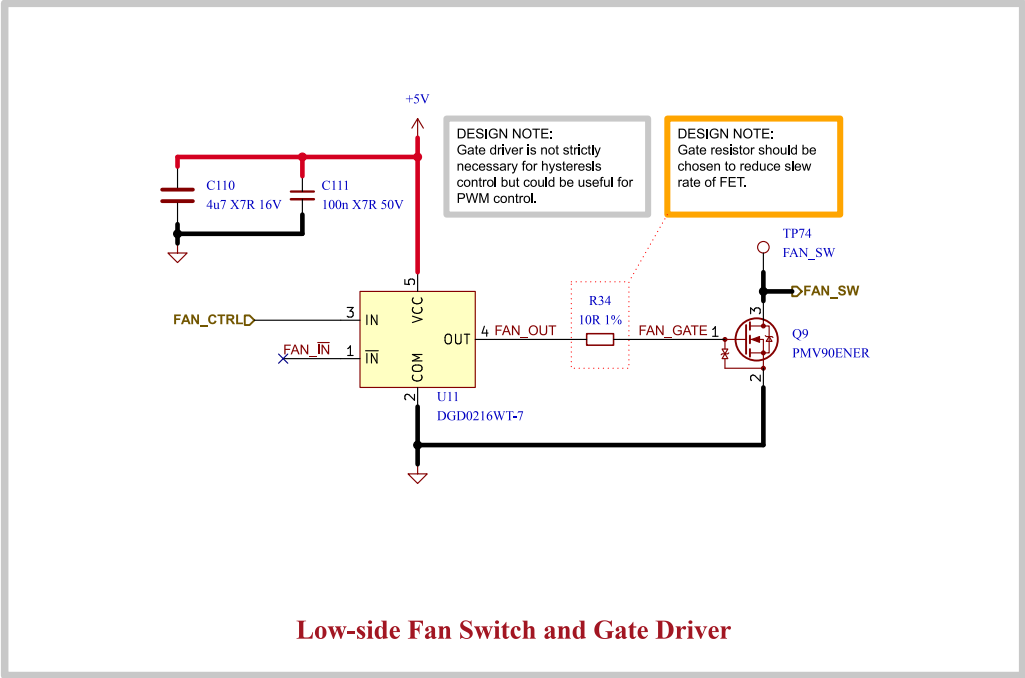
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[16] Interface - FD-CAN



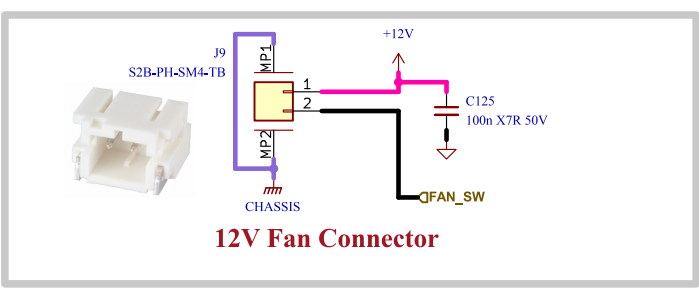
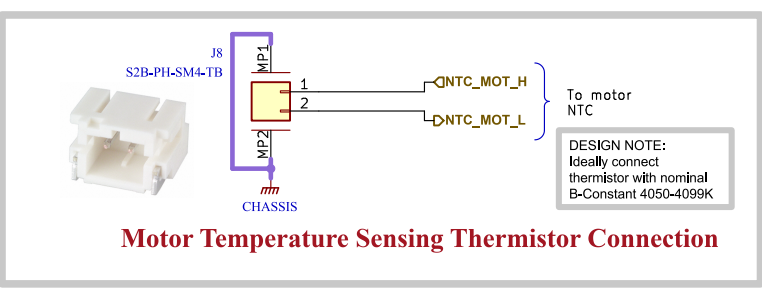
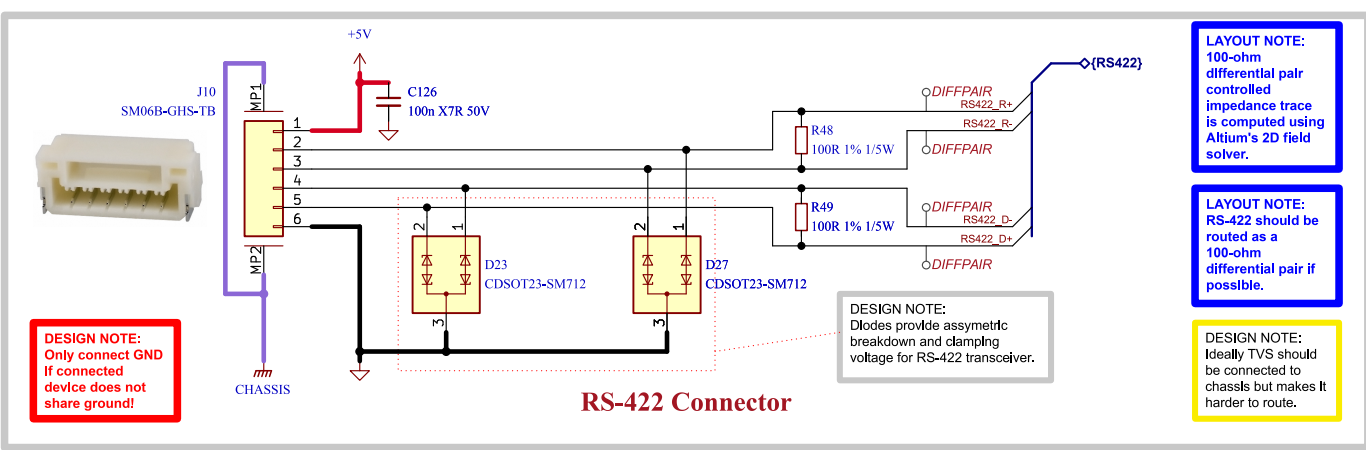
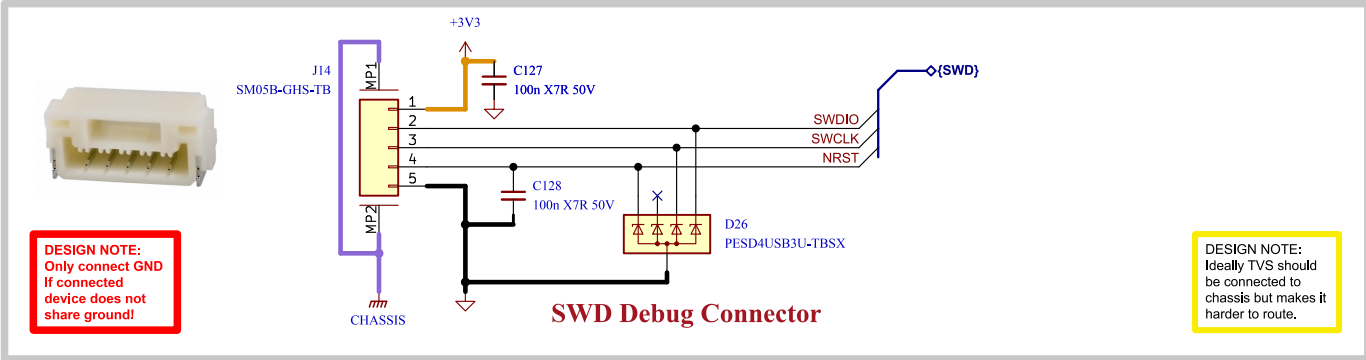
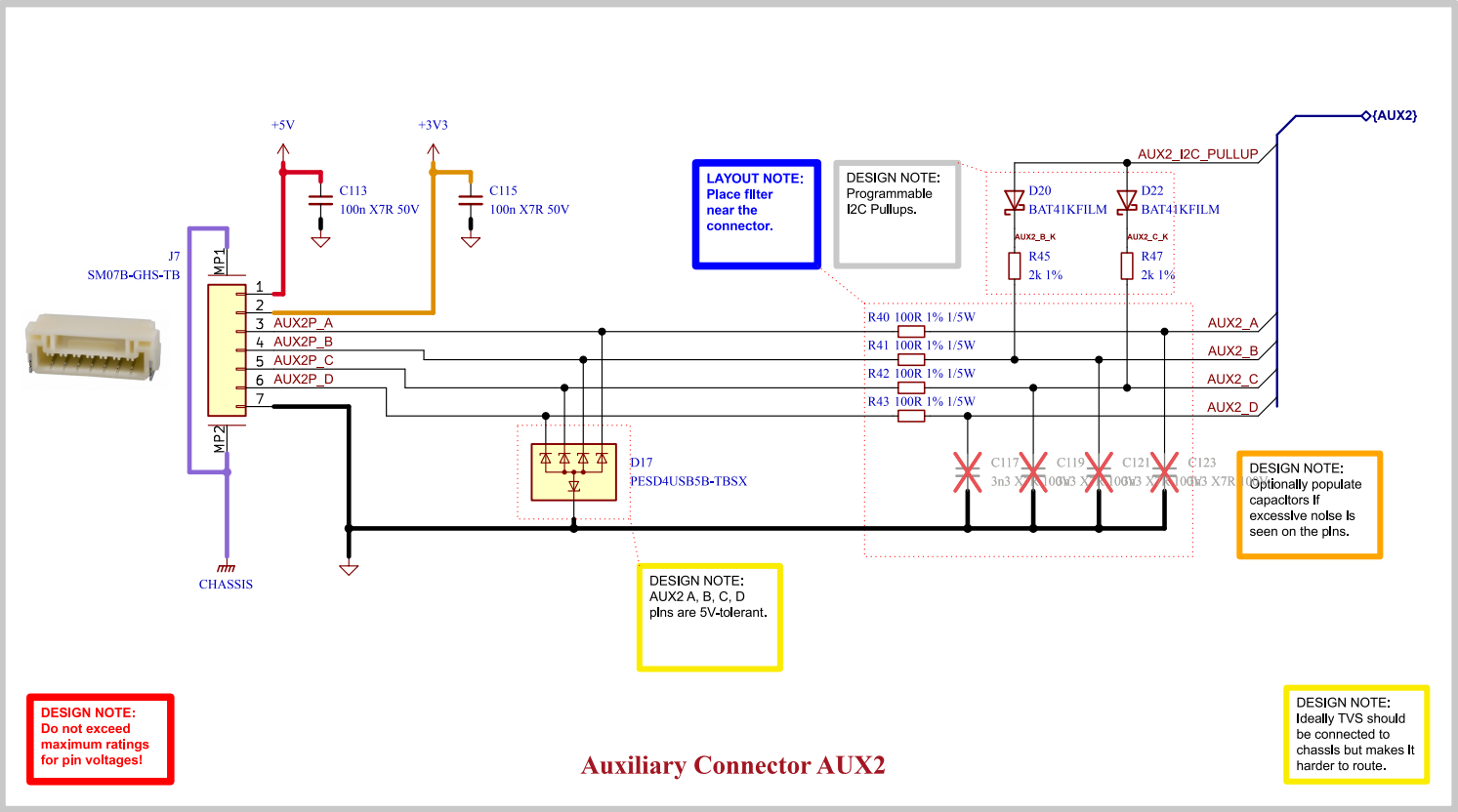
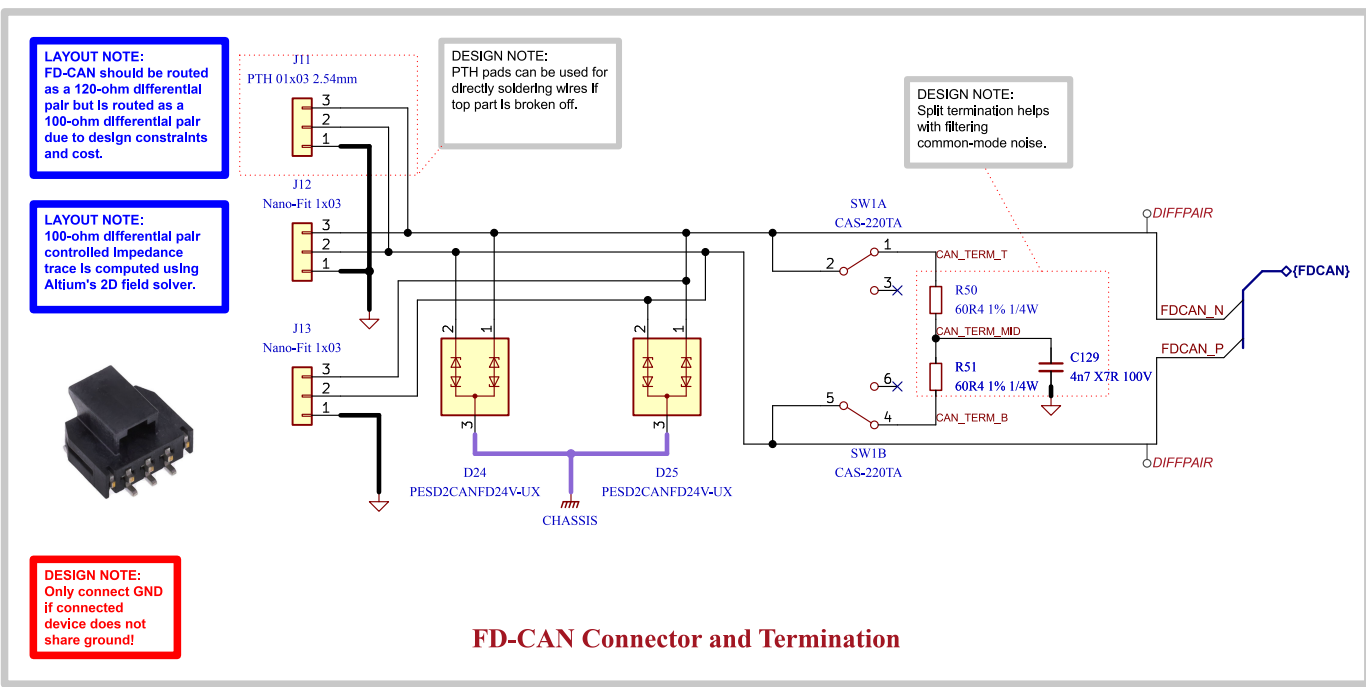
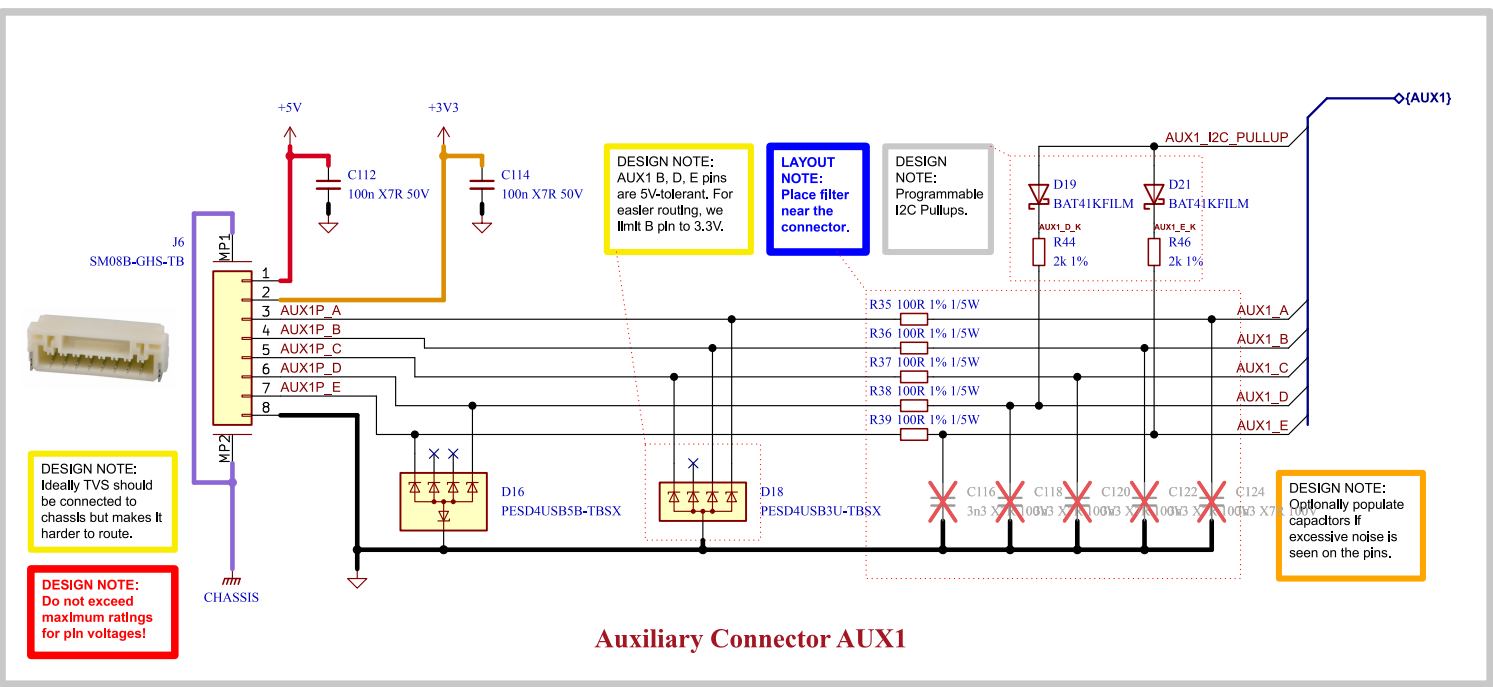
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[17] Interface - Fan Control



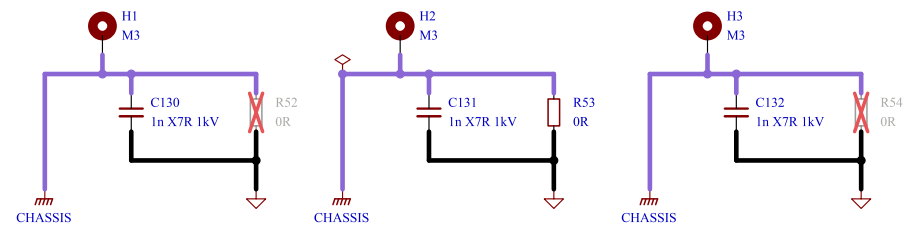
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	Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:	Size: A4	Sheet: 17 of 21

[18] Interface - Interconnects



Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research		Variant: Released	
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
	Sheet Path: /Project Architecture/Interface - Interconnects/	Reviewer:	Size: A3	Revision: 1.1
			Sheet: 18 of 21	

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis
grounding. Capacitors offer multi-point
low-impedance connections at
high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance
bond between chassis and signal GND.
Be aware of ground loops through cable
shield.

Mounting Holes and Chassis Connection

Front



Back



Fiducial Markers

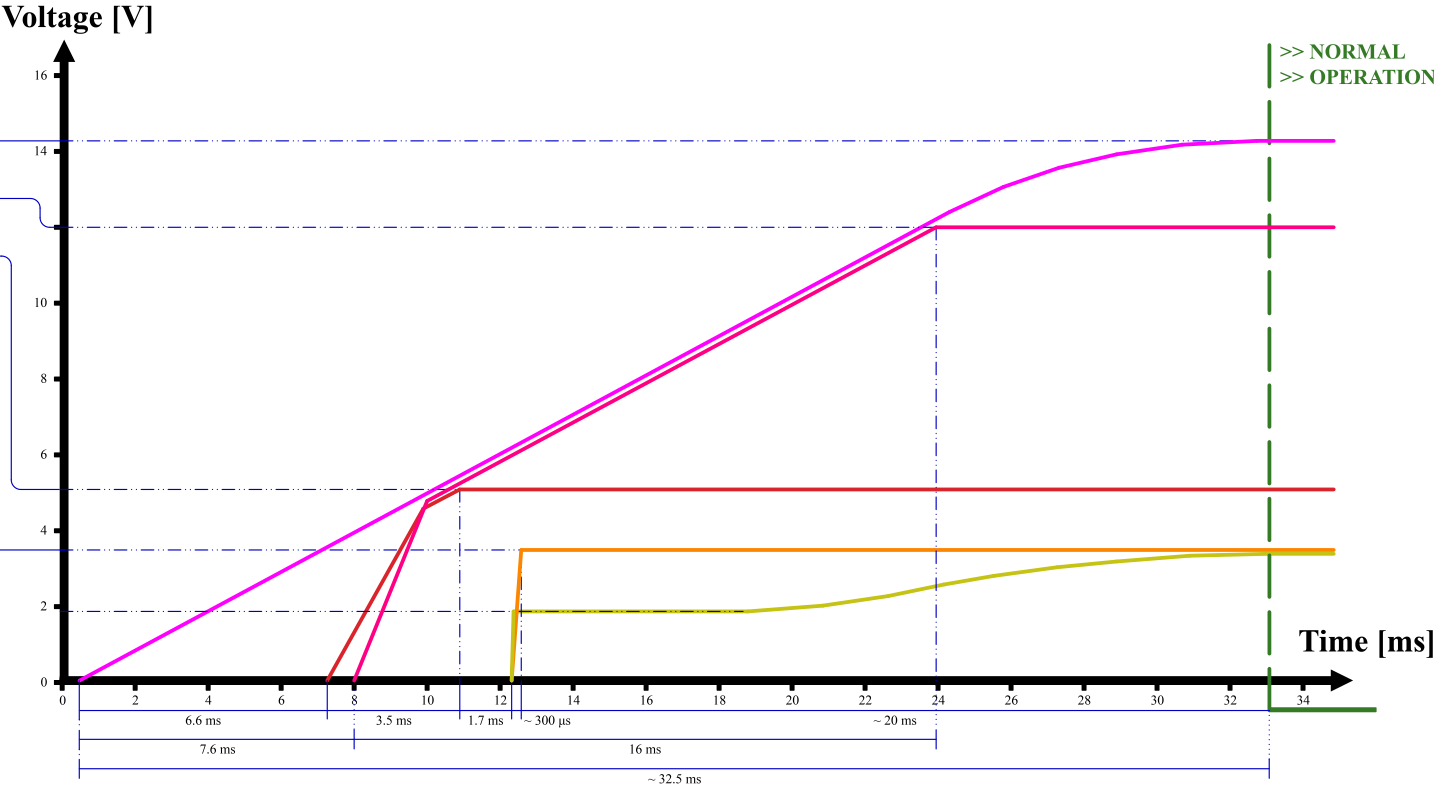
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	Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-22	Revision: 1.1	
	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21	

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V $\pm$ 1.5%
+5V	LMR36506	5V $\pm$ 1.5%
+3V3	TPS62172	3.3V $\pm$ 3%
+A3V3	LP2992	3.3V $\pm$ 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research	Variant: Released	
		Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12
	Sheet Path: /Power - Sequencing/	Reviewer:	Size: A4	Revision: 1.1

1	2	3	4	5									
[21] Revision History													
12-DEC-2023 - Initial Release Variant: v1.0 Preliminary - Changed CPH-CPL capacitor to 47nF (gate driver). - Changed FD-CAN transceiver IC. - Changed FETs for top cooled variant. - Added TVS protection and termination switch to FD-CAN. - Added low-side switched 12V 600mA source for external fan. - Added LDO for analog supply. - Changed input power TVS diode to bidirectional and added one diode per connector. - Moved SOx low-pass filter to MCU section. Should be placed near MCU to avoid noise coupling into ADC lines. - Added second onboard I2C magnetic encoder for disambiguation. - Switched PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing. - Changed RS422 pinout on connector. - Added ESD protection to all interfaces. - Added overvoltage protection on thermistor ADC inputs. - Changed buck regulators to optimize for low noise. - Added Pi filters to inputs of buck regulators and MCU analog supply. - Added decoupling caps next to power pins of connectors.	25-JAN-2024 - First Revision Variant: v1.0 Checked - Added controller target specifications. - Replaced 5V 300mA buck converter with 600mA version. - Added credits to moteus on cover page. - Added optional RC-Snubber to power stage. - Increased chassis length to go around the board. - CAN and power TVS diodes now go to chassis. - Changed clearance between nets to respect IEC60664-1 where possible. - Rectified comment on precharge. - Changed power TVS diode reference designator from "U" to "D". - Replaced chassis-GND capacitor by 1nF 1kV.	12-MAR-2024 - First Revision Variant: v1.0 Released - Modified power sequencing graph according to experimental data.	13-APR-2024 - Second Revision Variant: v1.1 Released - Added RC snubber passive values. - Added more vias for VBUS and LMR36006 GND pads. - Changed board version voltage reference from +3V3 to +A3V3. - Changed motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.	<table> <tr> <td rowspan="12"> Comments: Company: EPFL Xplore Research Sheet Title: Revision History Sheet Path: /Revision History/ </td><td> Variant: Released Project Name: Chienpanzé </td></tr> <tr> <td colspan="2"> Board Name: Amulet Motion Controller File Name: Revision History.kicad_sch Designer: Vincent Nguyen </td><td> Date: 2024-01-03 Revision: 1.1 </td></tr> <tr> <td colspan="2"> Reviewer: Size: A4 Sheet: 21 of 21 </td><td></td></tr> </table>		Comments: Company: EPFL Xplore Research Sheet Title: Revision History Sheet Path: /Revision History/	Variant: Released Project Name: Chienpanzé	Board Name: Amulet Motion Controller File Name: Revision History.kicad_sch Designer: Vincent Nguyen		Date: 2024-01-03 Revision: 1.1	Reviewer: Size: A4 Sheet: 21 of 21		
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	Board Name: Amulet Motion Controller File Name: Revision History.kicad_sch Designer: Vincent Nguyen		Date: 2024-01-03 Revision: 1.1										
	Reviewer: Size: A4 Sheet: 21 of 21												
1	2	3	4	5									